NNN-F-800a
November 5, 1962

SUPERSEDING
Int. Fed. Spec. NNN-F-00800 (ARMY-Cm1C)
June 30, 1961, and
Fed. Spec. DD-F-825
November 2, 1954

FEDERAL SPECIFICATION

FUNNEL, SEPARATORY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers separatory funnels.

1.2 Classification.

1.2.1 *Types and capacities.* The separatory funnels shall be of the following types and capacities, as specified **(see 6.2)**.

Type I—Pear-shaped
Type II—Cylindrical

Capacities—30 milliliters (ml.)
60 ml.
125 ml.
250 ml.
500 ml.
1000 ml.
2000 ml.

Type III—For single drops
Capacity—60 ml.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

DD-G-541—Glass (Laboratory).
UU-T-111—Tape; Paper, Gummed (Sealing and Securing).
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-676—Boxes, Set-Up, Paperboard.
PPP-C-843—Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Docu-

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ments, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Service Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-M-14077—Molding Plastic, Polytetrafluorethylene (TFE-Fluoro Carbon Resin).

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

Commercial Standard CS-21 — Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical Ground Joints.

(Copies may be obtained from Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Glass.* Glass used in the manufacture of the separatory funnels shall conform to the requirements of type I, class A of DD-G-541.

3.1.2 *Polytetrafluoroethylene.* Polytetrafluoroethylene used in the manufacture of stopcocks shall conform to the requirements of class 1 of MIL-M-14077.

3.2 Stopcocks and stoppers. The glass or polytetrafluoroethylene stopcocks and the glass stoppers shall be ground or finished to standard taper dimensions in conformance with Commercial Standard CS-21.

Stopcocks shall make a leakproof fit with the funnels with which they are originally supplied when tested as specified in 4.6.1. Stopcock plugs shall be provided with a retaining device or rubber washer so that the plug is not accidentally dislodged.

3.3 Physical requirements

3.3.1 *Types I and II.* Types I and II separatory funnels shall conform to the dimensions as specified in table I.

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TABLE I. *Dimensions for types I and II*

Capacity* Milliliters (ml)	Stem dimensions			S/T Stopper No.	Diameter, holes in stopcock
	Inside diameter, millimeters (mm) minimum	Wall thickness, mm minimum	Length, mm + 20 mm		
30	5.5	1.5	55	9 or 13	2
60	5.5	1.5	55	13 or 16	2
125	5.5	1.5	55	13 or 22	2
250	5.5	1.5	55	16 or 22	2 or 4
500	5.5	1.5	55	19 or 27	4
1000	5.5	1.5	55	22 or 27	4
2000	7.5	1.5	55	27 or 38	6

*Tolerance in capacity shall be + 5% to + 35% in excess of listed capacity.

3.3.2 *Type III.* Type III separatory funnels shall be of 60-ml. capacity with a tolerance of + 5 percent to + 35 percent. The stem length shall be 220 mm. ± 10 mm. Separatory funnels shall be fitted with a number 13 or 16 stopper. The stopcock shall have a bore hole in the plug of 2 mm. diameter.

3.4 **Area for marking.** Each separatory funnel shall have an area on one side roughened by blasting or spot-enamel to provide for marking. The area shall show no change or removal of enamel, if used, and shall be satisfactory for marking when tested as specified in 4.6.2.

3.5 **Identification marking.** Each separatory funnel shall be permanently and legibly marked with the manufacturer's name or trademark.

3.6 **Workmanship.** Separatory funnels shall be symmetrical in shape and free from defects that impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the separatory funnels of the same type and capacity offered for acceptance at one time, which have been produced by one manu-

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facturer under essentially the same manufacturing conditions.

4.4 Sampling.

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For test.* A sample of three separatory funnels shall be taken at random from each lot of 100 funnels or fraction thereof. Each funnel in the sample shall be subjected to the leakage test in 4.6.1 and one-third of the sample for each of the following tests: 4.6.2.1, 4.6.2.2, and 4.6.2.3.

4.5 **Examination** Sample separatory funnels and the level A and level B preparation for delivery thereof shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Separatory funnels.*

Categories	*Defects*
Critical:	None defined
Major:	*AQL* 2.5 *percent defective*
101	Stopcock or stopper not as specified.
102	Dimensions or capacity not within tolerance.
103	Defective (broken, cracks, chips, etc.).
104	Unsymmetrical in shape.
105	Marking illegible, incorrect, or incomplete.

4.5.1.2 *Preparation for delivery (section 5).*

Categories	*Defects*
Critical:	None defined
Major:	*AQL* 1.5 *percent defective*
101	Unit, intermediate, or shipping containers not as specified.
102	Number of funnels per intermediate containers not as specified or indicated.
103	Unlike funnels intermediately packaged or packed.
104	Unit, intermediate, or shipping containers damaged.
105	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete.

4.6 **Tests.** Tests shall be conducted as follows:

4.6.1 *Leakage.* Stopcocks shall be tested with all lubricant removed from the barrel and plug. Plug shall be wet with water and inserted into the barrel in the open position and turned 90 degrees to a firm set. Funnel shall be filled to capacity and allowed to stand 30 minutes. No leakage shall occur either across the barrel or from the ends of the barrel. Tests shall be made only with the plug supplied with the barrel. When seating the plug, it must not be turned back and forth, since this may damage the ground surfaces.

4.6.2 *Area for marking.* Permanency of area for marking shall be tested as follows:

4.6.2.1 *Alkali test.* Immerse the funnel completely in normal sodium hydroxide at room temperature, cover and boil for ½ hour, then let stand for 1 hour. Remove

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the funnel and rinse thoroughly with distilled water. Dry by rubbing vigorously with a cloth.

4.6.2.2 *Acid test (hydrochloric).* Immerse the funnel completely in concentrated hydrochloric acid (sp. gr. 1.18) at room temperature, cover and boil for ½ hour, and let stand for 1 hour. Remove funnel and rinse thoroughly with distilled water. Dry by rubbing vigorously with a cloth.

4.6.2.3 *Acid test (sulfuric).* Immerse the funnel completely in concentrated sulfuric acid (sp. gr. 1.84) at room temperature. Let stand for 1 hour. Remove funnel and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.7 **Acceptance/rejection criteria.** If any separatory funnel in the sample fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Each separatory funnel, with stopper and stopcock secured in a manner to prevent disengagement from the funnel, shall be packaged in a box conforming to PPP-B-566, style I or II, type and class optional; PPP-B-636, type I, class 1; or PPP-B-676, type, class, and style optional. The box shall be provided with fiberboard blocking or sufficient cushioning material conforming to type II, class B, of PPP-C-843 to insure against shifting of or damage to the funnel. Box closure shall be secured with tape conforming to UU-T-111.

5.1.1.2 *Intermediate packaging.* The number of funnels specified (see 6.2), of the same type and capacity, shall be intermediately packaged in a box conforming to type I, class 1, of PPP-B-636. The boxes shall be closed in accordance with the appendix of the box specification.

5.1.2 *Level C.* Funnels shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Funnels shall be packed in a shipping container conforming to PPP-B-621, class 2; PPP-B-601, overseas type; or PPP-B-576, style B, class 2. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto. Containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto.

5.2.2 *Level B.* Funnels shall be packed in accordance with 5.2.1 except that shipping containers shall conform to PPP-B-621, class 1; PPP-B-601, domestic type; or PPP-B-576, style A, class 1. When specified (see 6.2), funnels shall be packed in fiberboard boxes conforming to PPP-B-636, type I or II, class 1, style RSC, minimum bursting strength 275 pounds per square inch.

5.2.3 *Level C.* Funnels shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules and regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or

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order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Separatory funnels, covered by this specification, are for general laboratory use.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Level of packaging and packing required.
(*c*) Type and capacity of funnel required.
(*d*) Number of funnels per intermediate container required.
(*e*) Fiberboard box level B pack required.

6.3 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 **Transportation description.** Transportation description applicable to this item is:

Glassware, chemical laboratory.
Carload minimum weight 18,000 pounds, subject to Rule 34, Uniform Freight Classification.

Glassware, laboratory.
Motor volume minimum weight 18,000 pounds.

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